

HW2

RJ Cass

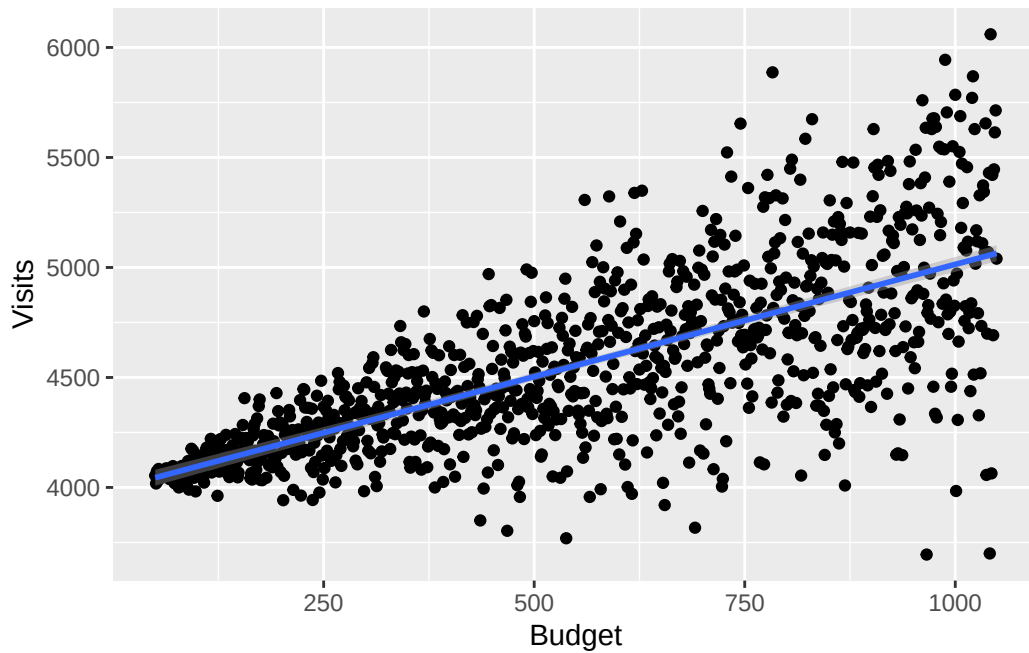
```
suppressPackageStartupMessages(library('ggplot2'))
suppressPackageStartupMessages(library('patchwork'))
suppressPackageStartupMessages(library('gamlss'))
suppressPackageStartupMessages(library('lmtest'))
suppressPackageStartupMessages(library('dplyr'))
web <- read.csv('website.csv')
```

HW 2.1

(a)

```
p11 <- ggplot(web, aes(x = Budget, y = Visits)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE)
p11
```

`geom\_smooth()` using formula = 'y ~ x'



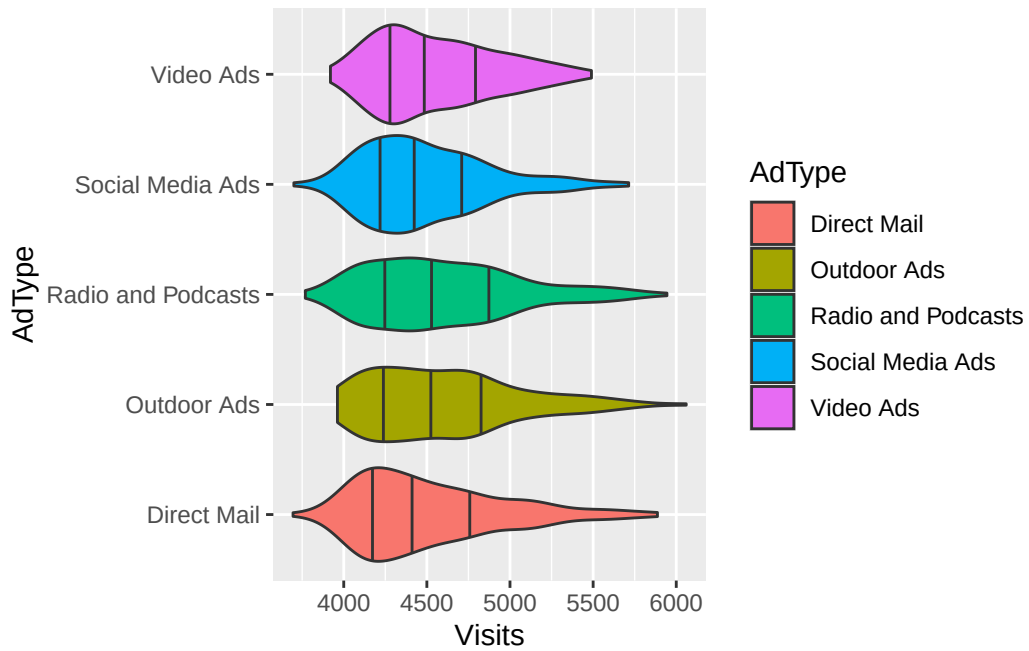
(b)

```
p12 <- ggplot(web, aes(x = Visits, y = AdType, fill = AdType)) +  
  geom_violin(draw_quantiles=c(0.25, .5, .75))
```

Warning: The `draw\_quantiles` argument of `geom\_violin()` is deprecated as of ggplot2 4.0.0.

i Please use the `quantiles.linetype` argument instead.

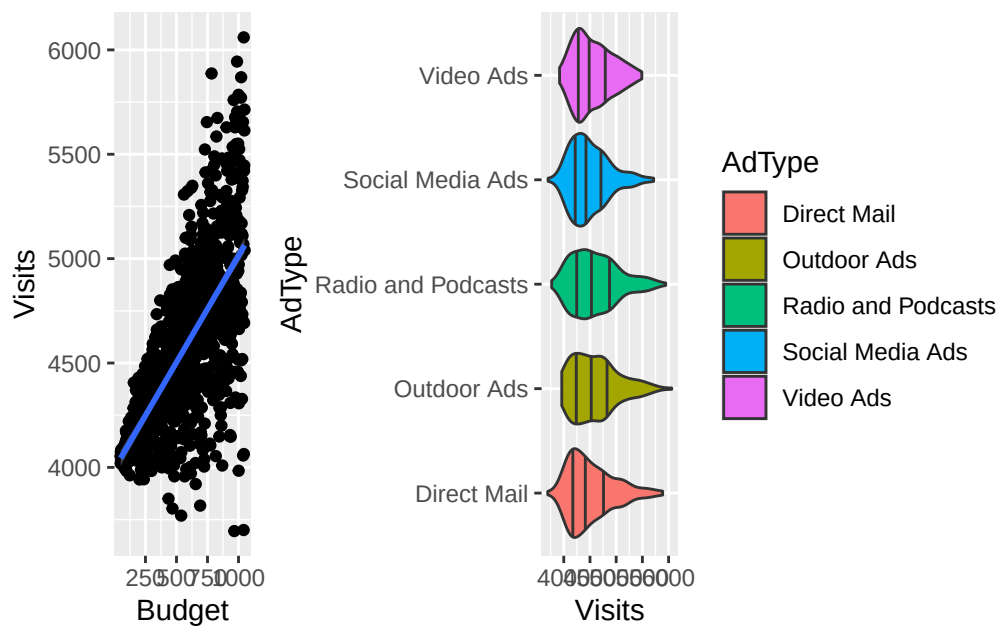
```
p12
```



Plotted together

p11 + p12

``geom_smooth()`` using formula = 'y ~ x'



(c)

```
mod_budget <- lm(Visits ~ Budget, data = web)
mod_adtype <- lm(Visits ~ AdType, data = web)
mod_full <- lm(Visits ~ Budget + AdType, data = web)

summary(mod_full)
```

Call:

```
lm(formula = Visits ~ Budget + AdType, data = web)
```

Residuals:

Min	1Q	Median	3Q	Max
-1313.54	-143.98	3.61	140.65	1095.48

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3995.4368	26.0962	153.104	<2e-16 ***
Budget	1.0167	0.0322	31.571	<2e-16 ***
AdTypeOutdoor Ads	17.6226	28.9569	0.609	0.543
AdTypeRadio and Podcasts	31.7844	29.0028	1.096	0.273
AdTypeSocial Media Ads	-40.2877	29.3655	-1.372	0.170
AdTypeVideo Ads	-10.3677	28.7366	-0.361	0.718

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 293 on 994 degrees of freedom

Multiple R-squared: 0.506, Adjusted R-squared: 0.5035

F-statistic: 203.6 on 5 and 994 DF, p-value: < 2.2e-16

```
anova(mod_budget, mod_full)
```

Analysis of Variance Table

Model 1: Visits ~ Budget

Model 2: Visits ~ Budget + AdType

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	998	85928668				
2	994	85343805	4	584863	1.703	0.1471

```
anova(mod_adtype, mod_full)
```

Analysis of Variance Table

Model 1: Visits ~ AdType

Model 2: Visits ~ Budget + AdType

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	995	170919701				
2	994	85343805	1	85575896	996.7	< 2.2e-16 ***

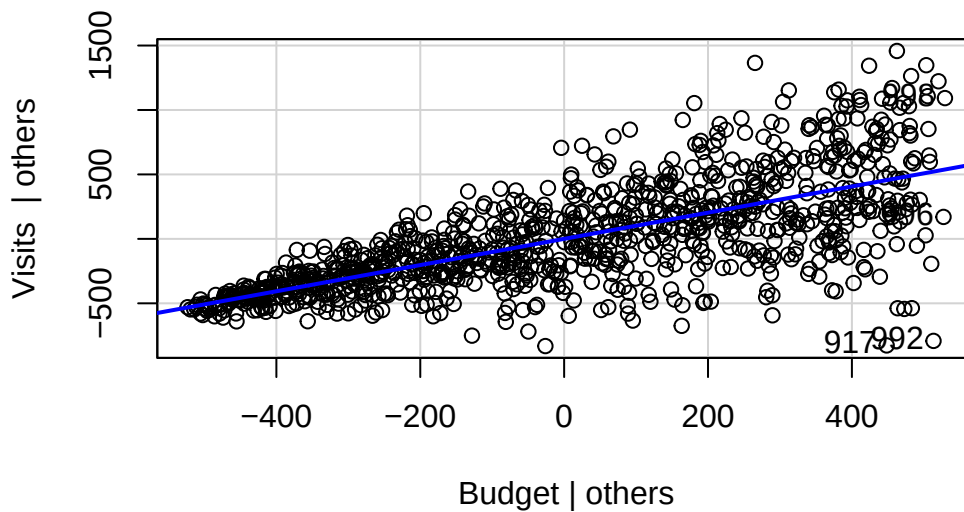
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

AdType does appear to be significant. Budget does not appear to be significant.

(d)

Linearity

```
car::avPlots(mod_full, terms = ~Budget)
```

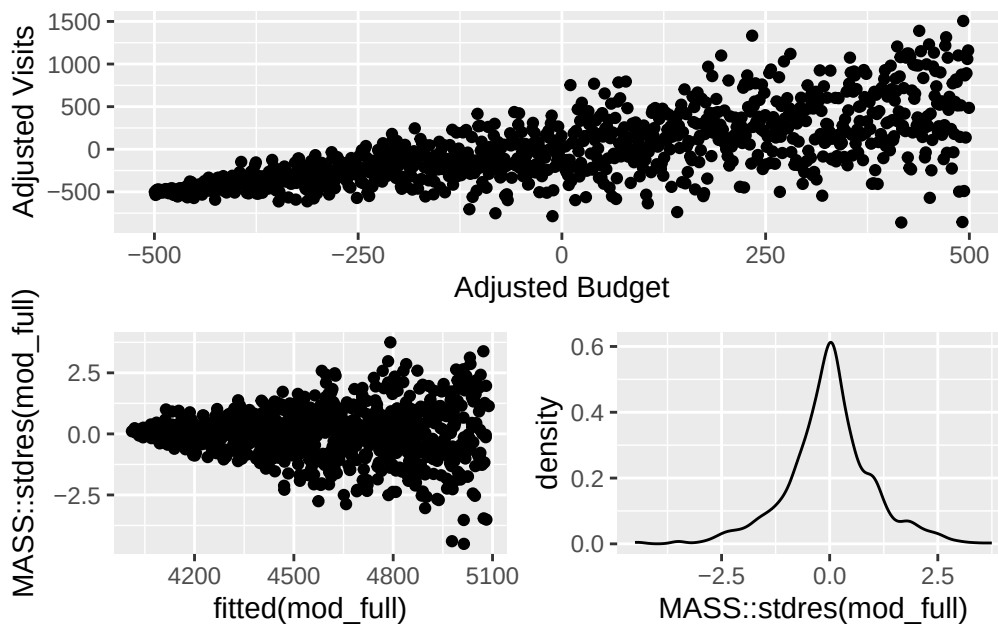


```

p141 <- ggplot(web, aes(x = (Budget - mean(Budget)), y = (Visits - mean(Visits)))) +
  geom_point() +
  labs(x = "Adjusted Budget", y = "Adjusted Visits")
p142 <- ggplot(web, aes(x = fitted(mod_full), y = MASS::stdres(mod_full))) +
  geom_point()
p143 <- ggplot(web, aes(x = MASS::stdres(mod_full))) +
  geom_density()

p141 / (p142 + p143)

```



(e)

```

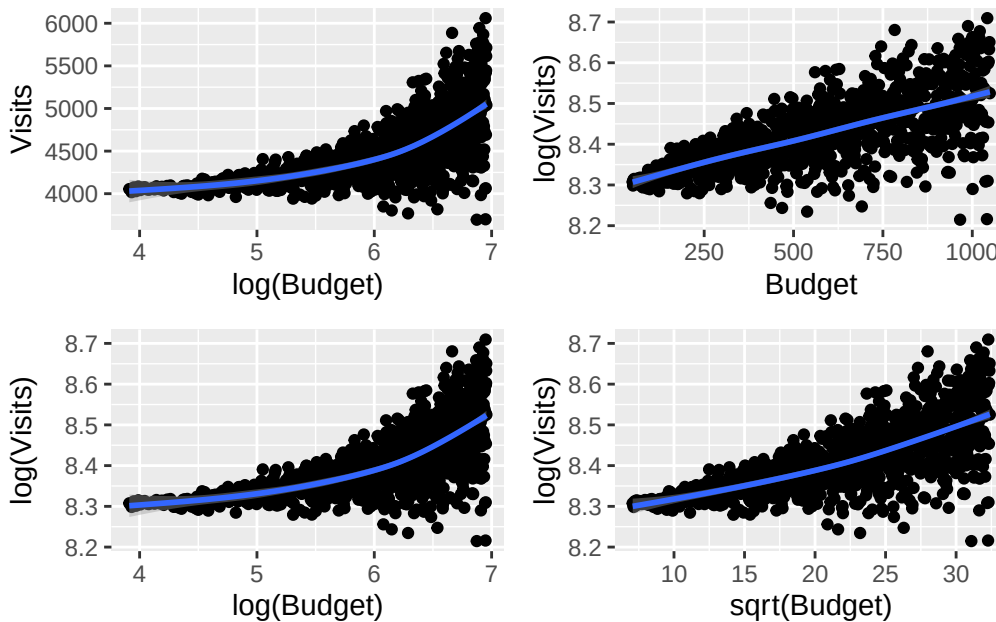
p151 <- ggplot(web, aes(x = log(Budget), y = Visits)) +
  geom_point() +
  geom_smooth()
p152 <- ggplot(web, aes(x = Budget, y = log(Visits))) +
  geom_point() +
  geom_smooth()
p153 <- ggplot(web, aes(x = log(Budget), y = log(Visits))) +
  geom_point() +
  geom_smooth()
p154 <- ggplot(web, aes(x = sqrt(Budget), y = log(Visits))) +

```

```
geom_point() +  
geom_smooth()
```

```
(p151 + p152) / (p153 + p154)
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'  
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'  
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'  
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



HW 2.2

(a)

```
gam_mod <- gamlss(Visits ~ Budget + AdType, sigma.formula = ~ Budget, data = web)
```

```
GAMLSS-RS iteration 1: Global Deviance = 13740.17  
GAMLSS-RS iteration 2: Global Deviance = 13724.39  
GAMLSS-RS iteration 3: Global Deviance = 13724.38  
GAMLSS-RS iteration 4: Global Deviance = 13724.38
```

Estimates for the β s

```
coef(gam_mod, what = "mu") # Mean model coefficients (betas)
```

	(Intercept)		Budget		AdTypeOutdoor Ads
	3994.3226521		1.0051834		8.1050678
AdTypeRadio and Podcasts		AdTypeSocial Media Ads		AdTypeVideo Ads	
	10.5937507		0.3614247		10.5669007

Estimates for θ_0 and θ_1

```
coef(gam_mod, what = "sigma") # Variance model coefficients
```

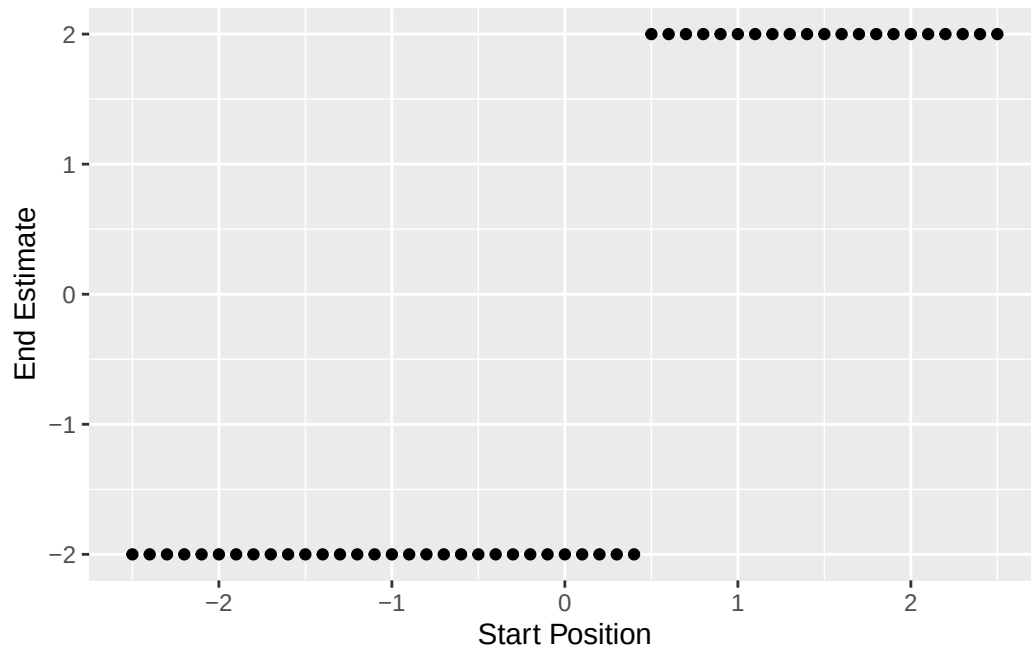
(Intercept)	Budget
4.336147892	0.002014751

(b)

I added in a grid to start over to demonstrate the change point where it starts converging to the different peaks.

```
multi_modal <- function(x) {  
  # Parameters for the first peak  
  mean1 <- -2  
  sd1 <- 1  
  peak1 <- dnorm(x, mean = mean1, sd = sd1)  
  
  # Parameters for the second peak  
  mean2 <- 2  
  sd2 <- .5  
  peak2 <- dnorm(x, mean = mean2, sd = sd2)  
  
  # Sum of both peaks  
  peak1 + peak2  
}  
  
start <- seq(-2.5, 2.5, by = .1)  
end <- unlist(lapply(start, \(x) optim(x, multi_modal, method = "L-BFGS-B", control = list(f  
  
ggplot(NULL, aes(x = start, y = end)) +
```

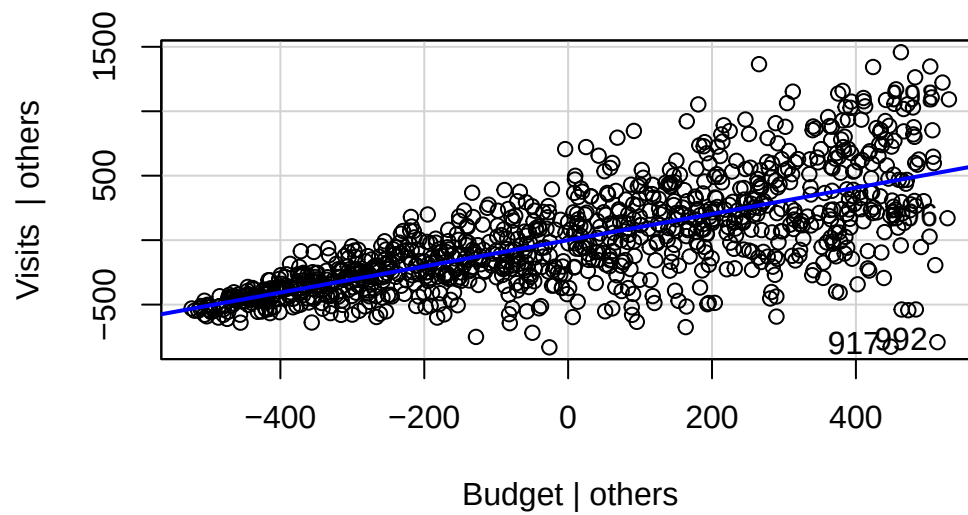
```
geom_point() +  
labs(x = "Start Position", y = "End Estimate")
```



HW 2.3

(a)

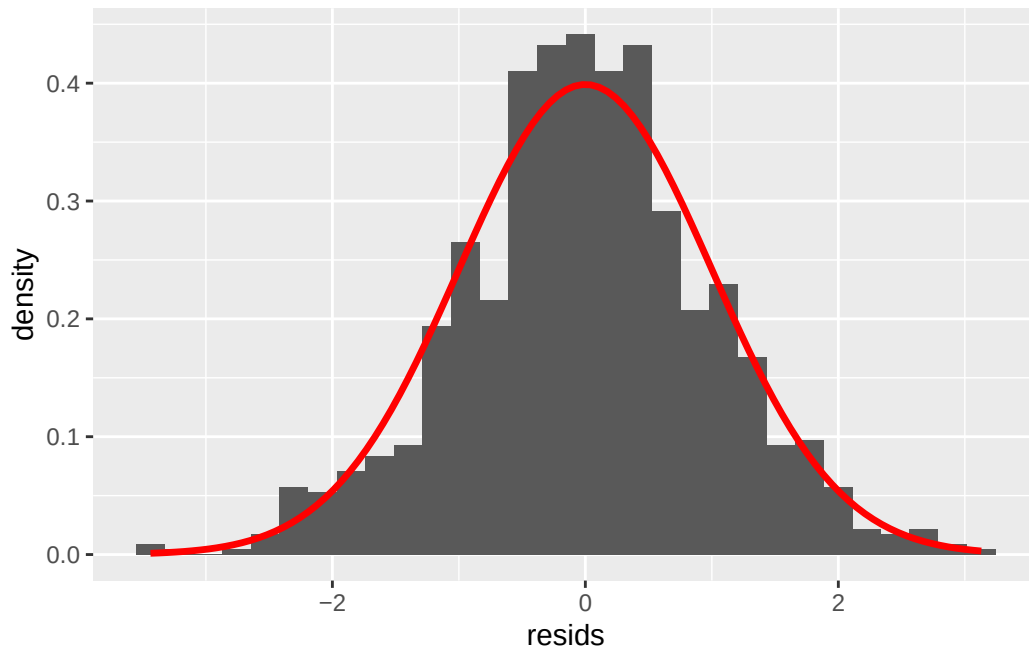
```
car::avPlots(mod_full, terms = ~Budget)
```



(b)

```
resids <- residuals(gam_mod)
resids <- as.data.frame(x = resids)
ggplot(data = resids, aes(x=resids)) +
  geom_histogram(aes(y = after_stat(density))) +
  stat_function(fun = dnorm, color = "red", linewidth = 1.2)
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
ks.test(resids$resids, "pnorm")
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data:  resids$resids
D = 0.031495, p-value = 0.2744
alternative hypothesis: two-sided
```

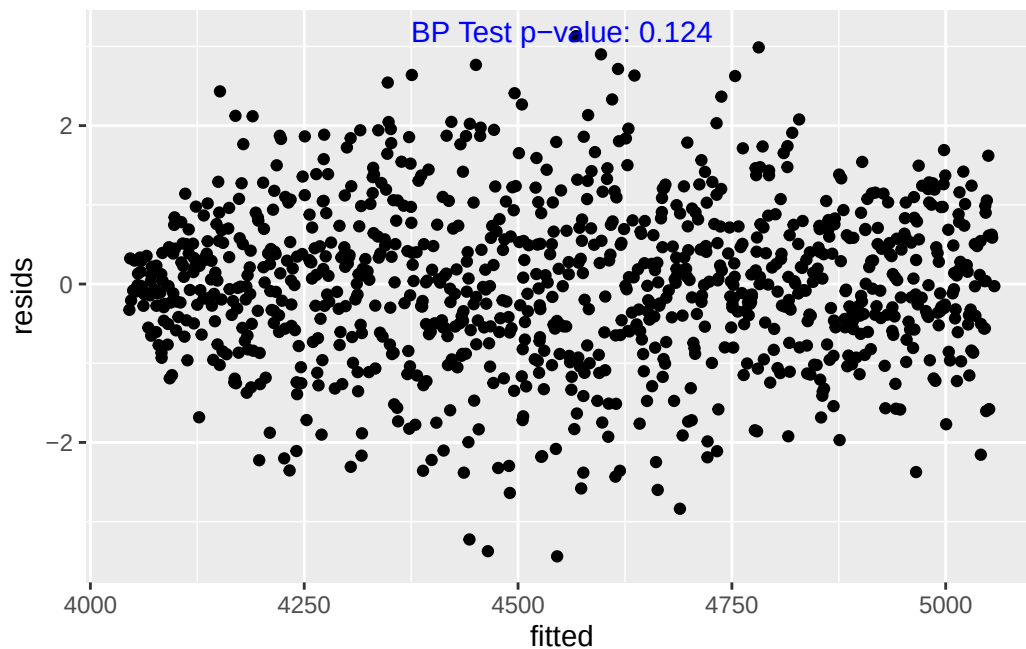
(c)

```
res_squared <- residuals(gam_mod)^2
aux_mod <- lm(res_squared ~ Budget + AdType, data = web)

bp <- bptest(aux_mod)
p_val <- bp$p.value
p_label <- paste("BP Test p-value:", format.pval(p_val, digits = 3))

fits <- fitted(gam_mod, what = "mu")
ggplot(NULL, aes(x = fits, y = resids$resids)) +
  geom_point() +
```

```
labs(x = "fitted", y = "resids") +
  annotate("text", x = median(fits), y = Inf, label = p_label, vjust = 1.5, color = "blue")
```



(d)

```
Rsq(gam_mod, type = c("Cox Snell"))
```

```
[1] 0.690614
```

(e)

```
cross_validate <- function(fold_num) {
  train_dt <- web[folds != fold_num, ]
  val_dt <- web[folds == fold_num, ]
  cv_mod <- gamlss(Visits ~ Budget + AdType, sigma.formula = ~ Budget, data = train_dt, tra
  preds <- predict(cv_mod, what = "mu", data = train_dt, newdata = val_dt, type = "respons
  raw_resids <- val_dt$Visits - preds
  rmse <- sqrt(mean(raw_resids^2))
  rmse
```

```

}

set.seed(1337)
K <- 25
folds <- rep(1:K, length = nrow(web)) %>% #Rep K folds n times
  sample()
## Using sapply()
rmse_results <- sapply(1:K, FUN=cross_validate)

## Plot the Results
ggplot() +
  geom_density(mapping=aes(x=rmse_results))

```

